

OPTIMAL FLOW TRANSPORT AND ITS ENTROPIC REGULARIZATION: A GPU-FRIENDLY MATRIX ITERATIVE ALGORITHM FOR FLOW BALANCE SATISFACTION

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Code: <https://github.com/asimfish/EOFT/>

ABSTRACT

The Sinkhorn algorithm, based on Entropic Regularized Optimal Transport (OT), has garnered significant attention due to its computational efficiency enabled by GPU-friendly matrix-vector multiplications. However, vanilla OT primarily deals with computations between the source and target nodes in a bipartite graph, limiting its practical application in real-world transportation scenarios. In this paper, we introduce the concept of Optimal Flow Transport (OFT) as an extension, where we consider a more general graph setting and the marginal constraints in vanilla OT are replaced by flow balance constraints. To obtain solutions, we incorporate entropic regularization into the OFT and introduce virtual flows for individual nodes to tackle the issue of potentially numerous isolated nodes lacking flow passages. Our proposition, the OFT-Sinkhorn algorithm, utilizes GPU-friendly matrix iterations to maintain flow balance constraints and minimize the objective function, and theoretical results for global convergence is also proposed in this paper. Furthermore, we enhance OFT by introducing capacity constraints on nodes and edges, transforming the OFT problem into a minimum-cost flow problem. We then present the Capacity-Constrained EOFT-Sinkhorn algorithm and compare it with the traditional Minimum cost flow (MCF) algorithm, showing that our algorithm is quite efficient for calculation. In particular, our EOFT-Sinkhorn is evaluated on high-precision and integer-precision MCF problems with different scales from one hundred to five thousand size, exhibiting significant time efficiency and the ability to approximate optimal solutions.

1 INTRODUCTION

Optimal Transport (OT) has shown increasingly important role in solving various problems in machine learning, including domain adaptation (Tzeng et al., 2017; Cui et al., 2018), generative models (Arjovsky et al., 2017; Li et al., 2022), self-supervised contrastive learning (Caron et al., 2020; Shi et al., 2023), and long-tail recognition (Peng et al., 2021; Shi et al., 2024b) etc. As an efficient method for solving OT, the Sinkhorn algorithm relies on matrix-vector iterations to solve the transportation problem, which is GPU-friendly with high-speed calculations. However, the Sinkhorn algorithm’s applicability is limited to bipartite graphs, as it only involves computations between sets of source and target nodes, which significantly deviates from real transportation scenarios. Naturally, a question arises: can optimal transport with matrix iterations be extended to more general graphs?

Numerous researchers have made significant efforts in the field of optimal transport on graphs. For instance, based on the graph structure, (Le et al., 2022) propose a variant of OT called Sobolev transport, which provides a closed-form solution for efficient computation. Additionally, in a related work, (Le et al., 2024) utilize a specific class of convex functions with an Orlicz structure to introduce

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